Rieffel-Deformed Completely Positive Maps: Full Answers to Questions 4.7 and 4.8

A candidate full solution to two questions in arXiv:2311.04630

23 July 2026

Status. Candidate full solution, likely valid, subject to human review. Both questions have affirmative answers for every locally compact abelian deformation group.

1 Source questions

Let $\Gamma \curvearrowright_\rho A$ be an action of a locally compact abelian group on a C^*-algebra and let Ψ be a continuous 2-cocycle on $\widehat{\Gamma}$. Skalski and Viselter construct, from every equivariant nondegenerate completely positive map $S : A \to A$, a completely positive map $S^\Psi : A^\Psi \to A^\Psi$ of norm one. They ask whether S^Ψ is always nondegenerate (Question 4.7), and whether an equivariant point-norm continuous semigroup $(S_t)_{t\geq 0}$ remains point-norm continuous after deformation (Question 4.8) [1].

Both questions appear on PDF page 26:

Transcription of the two live questions. Question 4.7 asks whether $S^\Psi : A^\Psi \to A^\Psi$ is always nondegenerate. Question 4.8 asks whether $(S_t^\Psi)_{t\geq 0}$ is point-norm continuous when $(S_t)_{t\geq 0}$ is a point-norm continuous semigroup of equivariant nondegenerate completely positive maps with $S_0 = \mathrm{id}_A$. Question 4.9 is not a live target here: immediately after stating it, the source reports an affirmative solution by Shalit and Skeide.

2 Result

Theorem 1. *In the full generality of Questions 4.7 and 4.8:*

1. $S^\Psi : A^\Psi \to A^\Psi$ *is nondegenerate for every equivariant nondegenerate completely positive map* $S : A \to A$*;*

2. *if* $(S_t)_{t\geq 0}$ *is a point-norm continuous semigroup as in Question 4.8, then* $(S_t^\Psi)_{t\geq 0}$ *is point-norm continuous.*

Thus both questions have affirmative answers for every locally compact abelian deformation group Γ.

3 Proof intuition

Put $B = A \rtimes_\rho \Gamma$ and write $\alpha = \widehat{\rho}^\Psi$ for the deformed dual action. The source proves that the crossed-product extension $\widetilde{S} : B \to B$ is nondegenerate, commutes with α, and restricts to S^Ψ on the Landstad algebra.

particular for universal and reduced crossed products.

We were not able to answer the following pertinent question:

Question 4.7. *In Proposition 4.5, is it always true that the restriction $S^\Psi : A^\Psi \to A^\Psi$ is non-degenerate?*

Our main aim in this paper is to deform *quantum convolution semigroups*, which will be achieved in Theorem 4.15. To deduce the required continuity it is tempting to try to use directly C*-algebraic arguments; this would be possible if we could provide a positive answer to the following question.

Question 4.8. *Suppose that $(S_t)_{t\geq 0}$ is a point–norm continuous semigroup of non-degenerate completely positive maps on A with $S_0 = \mathrm{id}_A$, commuting with the action $\Gamma \overset{\rho}{\curvearrowright} A$, i.e. such that $\rho_\gamma \circ S_t = S_t \circ \rho_\gamma$ for each $t \geq 0$, $\gamma \in \Gamma$. Is it then true that the deformed semigroup $(S_t^\Psi)_{t\geq 0}$ of completely positive maps on A^Ψ is point–norm continuous?*

In the specific context of Theorem 4.15 below we could answer Question 4.8 affirmatively with the aid of [SkV$_1$, Lemma 2.3], without appealing to von Neumann algebras as in Lemma 4.4, if we knew the following very general question had an affirmative answer.

Question 4.9. *Let $(S_t)_{t\geq 0}$ be a C_0-semigroup of non-degenerate completely positive maps on a C*-algebra A. Is it then true that the semigroup on $\mathrm{M}(A)$ obtained by canonically extending each operator S_t is point–strict continuous?*

We have been informed by Orr M. Shalit and Michael Skeide that they were recently able to solve Question 4.9 affirmatively in [ShS].

Figure 1: Questions 4.7–4.9 in [1, p. 26].

The strict-topology obstacle disappears after crossing one more time. Landstad duality also writes

$$B = A^\Psi \rtimes_{\rho^\Psi} \Gamma, \qquad \widetilde{S} = S^\Psi \rtimes \Gamma.$$

Crossing $\widetilde{S}$ by the dual action preserves nondegeneracy. Takai duality identifies the resulting double-crossed map with $S^\Psi \otimes \mathrm{id}_{\mathcal{K}(L^2\Gamma)}$. Nondegeneracy of a stabilization reflects back to the original completely positive map. This proves Question 4.7.

For Question 4.8, the extensions $\widetilde{S}_t$ form a point-norm continuous semigroup on B: this follows first on $C_c(\Gamma, A)$ from the L^1 estimate and then by density. The Landstad algebra is the generalized fixed-point algebra of the weakly proper $\widehat{\Gamma} \rtimes \widehat{\Gamma}$-algebra (B, α, λ). Its dense compactly supported core consists of averages

$$E\big(\lambda(f) b \lambda(g)\big), \qquad f, g \in C_c(\widehat{\Gamma}),\ b \in B.$$

The averaging map is norm bounded when f and g are fixed. Since every $\widetilde{S}_t$ fixes $\lambda(C_0(\widehat{\Gamma}))$ and commutes with α, continuity on B passes to this dense core, and contractivity passes it to all of A^Ψ.

4 Crossed products reflect nondegeneracy

We use the standard crossed-product extension of an equivariant completely positive map: if $R : C \to C$ commutes with an action $G \curvearrowright_\beta C$, then

$$(R \rtimes G)(h)(s) = R(h(s)), \qquad h \in C_c(G, C).$$

The source recalls that this extension exists without assuming nondegeneracy, and that it is nondegenerate whenever R is nondegenerate [1, Proposition 4.5 and Remark 4.6].

Lemma 2 (Takai naturality for completely positive maps)**.** *Let G be a locally compact abelian group, let $G \curvearrowright_\beta C$, and let $R : C \to C$ be an equivariant completely positive map. Under the Takai isomorphism*

$$\Phi_\beta : (C \rtimes_\beta G) \rtimes_{\widehat{\beta}} \widehat{G} \longrightarrow C \otimes \mathcal{K}(L^2 G),$$

one has

$$\Phi_\beta \circ \big((R \rtimes G) \rtimes \widehat{G}\big) = (R \otimes \mathrm{id}_{\mathcal{K}(L^2G)}) \circ \Phi_\beta.$$

Proof. The usual Takai transform is defined first on $C_c(\widehat{G}, C_c(G, C))$. Its integral-kernel formula is linear in the C-valued coefficients and uses only scalar Fourier transforms and β-translates of those coefficients. The double-crossed extension applies R to each coefficient. Since $R\beta_s = \beta_s R$, applying R before or after the Takai transform gives the same kernel, with R acting on its C-entry and the identity acting on its compact-operator entry. The displayed identity follows on the dense double-crossed algebra and hence everywhere by boundedness. This is the naturality of the standard Takai duality isomorphism [3]. □

Lemma 3 (Stable reflection)**.** *Let $R : C \to C$ be completely positive. If $R \otimes \mathrm{id}_{\mathcal{K}(H)}$ is nondegenerate for a nonzero Hilbert space H, then R is nondegenerate.*

Proof. Let (e_i) be a positive contractive approximate identity of C and let (p_j) be the net of finite-rank projections on H, directed by inclusion of their ranges. Then $(e_i \otimes p_j)_{(i,j)}$ is an approximate identity of $C \otimes \mathcal{K}(H)$. Fix $c \in C$ and a rank-one projection p on H. Nondegeneracy of $R \otimes \mathrm{id}$ gives

$$\big(R(e_i) \otimes p_j\big)(c \otimes p) \longrightarrow c \otimes p.$$

After taking j with $p_j p = p$, this says $\|R(e_i)c - c\| \to 0$. Multiplication on the other side similarly gives $\|cR(e_i) - c\| \to 0$. Thus $R(e_i) \to 1$ strictly in $\mathrm{M}(C)$. □

Proposition 4 (Crossed products reflect nondegeneracy)**.** *Let G be a locally compact abelian group and let $R : C \to C$ be an equivariant completely positive map. If $R \rtimes G$ is nondegenerate, then R is nondegenerate.*

Proof. The map $R \rtimes G$ commutes with the dual action. Since it is nondegenerate, its crossed-product extension

$$(R \rtimes G) \rtimes \widehat{G}$$

is nondegenerate by the crossed-product preservation result quoted above. Lemma 2 transports this map to $R \otimes \mathrm{id}_{\mathcal{K}(L^2G)}$, which is therefore nondegenerate. Lemma 3 now implies that R is nondegenerate. □

5 Continuity on a generalized fixed-point algebra

Lemma 5 (Continuity of crossed-product semigroups). *Let a locally compact group G act on C. If $(R_t)_{t\geq 0}$ is a point-norm continuous semigroup of equivariant completely positive contractions, then $(R_t \rtimes G)_{t\geq 0}$ is point-norm continuous on $C \rtimes G$.*

Proof. For $h \in C_c(G, C)$,

$$\|(R_t \rtimes G)(h) - h\|_{C\rtimes G} \leq \int_G \|R_t(h(s)) - h(s)\|\, ds.$$

The range of h is norm compact. Pointwise convergence $R_t(c) \to c$ and contractivity imply uniform convergence on norm-compact subsets of C. The right-hand side therefore tends to zero as $t \downarrow 0$. Since $C_c(G, C)$ is dense in the crossed product and the maps are contractions, the conclusion extends to every element. The semigroup law then gives continuity at every $t \geq 0$. □

Lemma 6 (Averaging-core transfer). *Let (B, α, ϕ) be a weakly proper $G \rtimes G$-algebra, where G acts on itself by translation and $\phi : C_0(G) \to \mathrm{M}(B)$ is the structural map. Let $(T_t)_{t\geq 0}$ be nondegenerate completely positive contractions such that:*

1. *$t \mapsto T_t(b)$ is norm continuous for every $b \in B$;*
2. *T_t commutes with α;*
3. $$T_t\big(\phi(f)b\phi(g)\big) = \phi(f)T_t(b)\phi(g)$$
   for $f, g \in C_c(G)$ and $b \in B$.

Then the restrictions of T_t to the generalized fixed-point algebra $\mathrm{Fix}(B)$ form a point-norm continuous semigroup.

Proof. Put $B_c = \phi(C_c(G))B\phi(C_c(G))$. For the proper translation action, the averaging map

$$E(a) = \int_G^{\mathrm{st}} \alpha_s(a)\, ds$$

maps B_c onto a dense subalgebra of $\mathrm{Fix}(B)$. More precisely, for fixed $f, g \in C_c(G)$ the map

$$E_{f,g} : B \longrightarrow \mathrm{M}(B), \qquad b \longmapsto E(\phi(f)b\phi(g))$$

is norm continuous [2, Lemma 2.5(4)], and the span of its ranges is dense in the generalized fixed-point algebra [2, Proposition 2.2 and Definition 2.8].

Nondegeneracy gives a strictly continuous extension of T_t on bounded subsets of $\mathrm{M}(B)$. Equivariance and the bimodule identity therefore permit T_t to pass through the strict averaging integral:

$$T_t(E(\phi(f)b\phi(g))) = E(\phi(f)T_t(b)\phi(g))\,.$$

Consequently, norm continuity of $T_t(b)$ and boundedness of $E_{f,g}$ give point-norm continuity on the dense averaging core. Contractivity extends it to $\mathrm{Fix}(B)$. □

6 Proof of the main theorem

Proof of Theorem 1. Set

$$B = A \rtimes_\rho \Gamma, \qquad \alpha = \widehat{\rho}^\Psi.$$

Proposition 4.5 of the source supplies a nondegenerate completely positive map $\widetilde{S} : B \to B$ which commutes with α, fixes the canonical copy $\lambda(C_0(\widehat{\Gamma}))$, and restricts to S^Ψ on A^Ψ [1].

Landstad duality gives a canonical identification

$$B = A^\Psi \rtimes_{\rho^\Psi} \Gamma$$

which preserves the canonical copies of A^Ψ and Γ. Under this identification, the multiplicative-domain formula from Proposition 4.5 gives

$$\widetilde{S} = S^\Psi \rtimes \Gamma$$

on $C_c(\Gamma, A^\Psi)$. Since $\widetilde{S}$ is nondegenerate, Proposition 4 shows that S^Ψ is nondegenerate. This answers Question 4.7.

Now let $(S_t)_{t\geq 0}$ be as in Question 4.8 and let $\widetilde{S}_t = S_t \rtimes \Gamma$ on B. By Lemma 5, $(\widetilde{S}_t)$ is point-norm continuous on B. The triple

$$(B, \widehat{\rho}^\Psi, \lambda : C_0(\widehat{\Gamma}) \to \mathrm{M}(B))$$

is a weakly proper $\widehat{\Gamma} \rtimes \widehat{\Gamma}$-algebra, and its generalized fixed-point algebra is precisely the Landstad algebra A^Ψ [2, Theorem 4.6]. Proposition 4.5 of the source gives the equivariance and bimodule identities required in Lemma 6. That lemma therefore shows that the restrictions S_t^Ψ form a point-norm continuous semigroup on A^Ψ. This answers Question 4.8. □

7 Verification and novelty status

The proof has three independent consistency checks.

- If Γ is discrete, $\widehat{\Gamma}$ is compact and the averaging-core proof reduces to the compact fixed-point argument in the earlier partial packet.
- If the cocycle is trivial, the Landstad algebra is the undeformed algebra and both conclusions reduce to their hypotheses.
- No conclusion is drawn from convergence in the ambient strict topology. Question 4.7 is transferred to a stabilization, while Question 4.8 is proved in norm on a dense generalized fixed-point core.

A bounded novelty search on 23 July 2026 used the exact wording of Questions 4.7 and 4.8, the source title and authors, citation searches for arXiv:2311.04630, and combinations of "Takai duality", "crossed-product completely positive map", "generalized fixed-point algebra", "nondegenerate", and "point-norm continuous". No paper explicitly answering either question, and no prior statement of the two arguments above, was found. Buss and Echterhoff's paper predates the questions and supplies the averaging-core machinery, but does not state these consequences. This is a bounded search, not an exhaustive novelty certification.

Human-review recommendation. Review Proposition 4, especially Takai naturality for completely positive maps, and the identification of the Landstad algebra with the generalized fixed-point completion used in Lemma 6. These are the only two non-elementary bridges in the proof.

References

[1] A. Skalski and A. Viselter, *Convolution semigroups on Rieffel deformations of locally compact quantum groups*, arXiv:2311.04630; Lett. Math. Phys. **114** (2024), 52.

[2] A. Buss and S. Echterhoff, *Universal and exotic generalized fixed-point algebras for weakly proper actions and duality*, arXiv:1304.5697.

[3] S. Imai and H. Takai, *On a duality for C^*-crossed products by a locally compact group*, J. Math. Soc. Japan **30** (1978), 495–504.